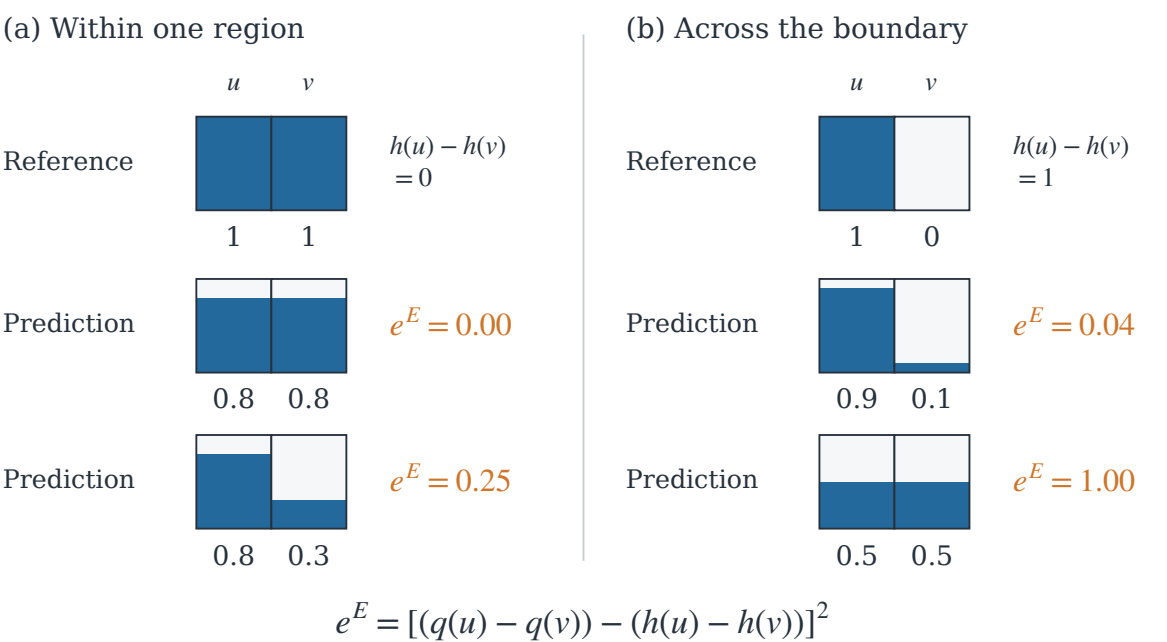

$$e^E = [(q(u) - q(v)) - (h(u) - h(v))]^2$$